

Duality of LPP

Primal Problem: The canonical form of the given problem is called the primal problem.

Consider the maximization problem:

$$\left. \begin{array}{l} \text{Max } Z = C^T x \\ \text{Subject to: } Ax \leq b, \ x \geq 0. \end{array} \right\} \text{Primal problem.}$$

Dual Problem: The corresponding dual of the above problem is

$$\left. \begin{array}{l} \text{Min } Z = b^T y \\ \text{Subject to: } A^T y \geq C, \ y \geq 0. \end{array} \right\} \text{Dual problem.}$$

Together, these two problems constitute a primal dual pair. Here x is called primal variable and y is called dual variable.

Similarly, for minimization problem:

$$\left. \begin{array}{l} \text{Min } Z = C^T x \\ \text{Subject to: } Ax \leq b, \ x \geq 0. \end{array} \right\} \text{Primal problem.}$$

Dual Problem: The corresponding dual of the above problem is

$$\left. \begin{array}{l} \text{Max } Z = b^T y \\ \text{Subject to: } A^T y \geq C, \ y \geq 0. \end{array} \right\} \text{Dual problem.}$$

Observations

- For every primal constraint, there is a dual variable. So, the number of dual variables is exactly equal to the number of primal constraints.
- For every primal variable, there is a dual constraint. So, the number of dual constraints is exactly equal to the number of primal variables.
- The columns in the primal coefficient matrix become the rows in the dual coefficient matrix.
- The objective coefficients of the primal variables become the right-hand side constants of the dual constraints, whereas the right-hand side constants of the primal constraints become the objective coefficients of the dual problem.

Ex: Find the dual of the given linear programming problem:

$$\text{Min } Z = 2x_1 + 4x_2 + 6x_3$$

$$x_1 - x_2 \leq 4,$$

$$\text{Subject to: } 2x_1 + x_2 + x_3 \geq 5, \quad x_1, x_2, x_3 \geq 0$$

$$x_1 + 4x_2 - 3x_3 \geq 7,$$

Solution: To get the primal problem, we rewrite the given LPP in canonical form:

$$\text{Min } Z = 2x_1 + 4x_2 + 6x_3$$

$$-x_1 + x_2 \geq -4,$$

$$\text{Subject to: } 2x_1 + x_2 + x_3 \geq 5, \quad x_1, x_2, x_3 \geq 0$$

$$x_1 + 4x_2 - 3x_3 \geq 7,$$

We consider y_1, y_2, y_3 three dual variables corresponding to the three constraints of the primal problem.

Therefore the dual is:

$$\text{Max } Z' = -4y_1 + 5y_2 + 7y_3$$

Subject to:

$$-y_1 + 2y_2 + y_3 \leq 2,$$

$$y_1 + y_2 + 4y_3 \leq 4, \quad y_1, y_2, y_3 \geq 0.$$

$$y_2 - 3y_3 \leq 6,$$

Relation between Primal constraints and corresponding dual variables:

- If i^{th} constraints of primal is " $\leq$ " type $\Leftrightarrow i^{th}$ variable of dual is " ≥ 0 " type.
- If i^{th} constraints of primal is " $\geq$ " type $\Leftrightarrow i^{th}$ variable of dual is " ≤ 0 " type.
- If i^{th} constraints of primal is " $=$ " type $\Leftrightarrow i^{th}$ variable of dual is unrestricted type.

Relation between Primal variables and corresponding dual constraints:

- If i^{th} variable of primal is unrestricted type $\Leftrightarrow i^{th}$ constraints of dual is " $=$ " type.
- If i^{th} variable of primal is " ≥ 0 " type $\Leftrightarrow i^{th}$ constraints of dual is " $\leq$ " type.
- If i^{th} variable of primal is " ≤ 0 " type $\Leftrightarrow i^{th}$ constraints of dual is " $\geq$ " type.

Remark: Dual of the dual is again the primal.

Given the following Linear Programming Problem (LPP), write the dual:

$$\text{Max } Z = 4x_1 + 6x_2$$

Subject to:

$$2x_1 + 3x_2 = 5,$$

$$x_1 + x_2 \leq 7, \quad x_1 \geq 0, x_2 \text{ is unrestricted in sign.}$$

Solution: Substitute $x_2 = x'_2 - x''_2$ in the given LPP and rewrite the LPP in canonical form:

$$\text{Max } Z = 4x_1 + 6x'_2 - 6x''_2$$

Subject to:

$$2x_1 + 3x'_2 - 3x''_2 \leq 5$$

$$-2x_1 - 3x'_2 + 3x''_2 \leq -5$$

$$x_1 + x'_2 - x''_2 \leq 7, \quad x_1, x'_2, x''_2 \geq 0.$$

Let y_1, y_2, y_3 are the dual variables. The dual problem is the following form:

$$\text{Min } Z' = 5y_1 - 5y_2 + 7y_3$$

Subject to:

$$2y_1 - 2y_2 + y_3 \geq 4$$

$$3y_1 - 3y_2 + y_3 \geq 6$$

$$-3y_1 + 3y_2 - y_3 \geq -6, \quad y_1, y_2, y_3 \geq 0.$$

The above LPP can be written as:

$$\text{Min } Z' = 5y_1 - 5y_2 + 7y_3$$

Subject to:

$$2y + y_3 \geq 4$$

$$3y + y_3 = 6, \quad y \text{ is unrestricted in sign and } y_3 \geq 0.$$

Weak Duality Theorem: *Let x^* be feasible solution to the primal problem:*

$$\max f(x) = C^T x$$

$$\text{Subject to: } Ax \leq b, \quad x \geq 0.$$

Where $x \in \mathbb{R}^n, C \in \mathbb{R}^n, b \in \mathbb{R}^m$ and A be a $m \times n$ real matrix. Let ω^ be a feasible solution to the dual problem:*

$$\min g(x) = b^T \omega$$

$$\text{Subject to: } A^T \omega \geq C, \quad \omega \geq 0,$$

where $\omega \in \mathbb{R}^m$. Then we have the inequality:

$$C^T x^* \leq b^T \omega.$$

Proof of Weak Duality Theorem:

Since, x^* is a feasible solution of the primal problem, it satisfies primal constraints:

$$Ax^* \leq b, \quad x^* \geq 0.$$

Taking transport in both side:

$$(x^*)^T A^T \leq b^T.$$

Post-multiply the above inequality by ω^* :

$$(x^*)^T A^T \omega^* \leq b^T \omega^* \tag{1}$$

Again, since ω^* be a feasible solution of the dual problem, it satisfies:

$$A^T \omega^* \geq C$$

Take transpose and post multiply by x^* on the above inequality:

$$(\omega^*)^T Ax^* \geq C^T x^* \tag{2}$$

Notice that $(x^*)^T A^T \omega^* = (\omega^*)^T Ax^*$, combine the relations (1) and (2), we get

$$C^T x^* \leq b^T \omega^*.$$

Corollary: *Let x^* be feasible solution to the primal problem:*

$$\max f(x) = C^T x$$

$$\text{Subject to: } Ax \leq b, \quad x \geq 0.$$

Where $x \in \mathbb{R}^n, C \in \mathbb{R}^n, b \in \mathbb{R}^m$ and A be a $m \times n$ real matrix. Let ω^ be a feasible solution to the dual problem:*

$$\min g(x) = b^T \omega$$

$$\text{Subject to: } A^T \omega \geq C, \quad \omega \geq 0,$$

where $\omega \in \mathbb{R}^m$. If $C^T x^ = b^T \omega^*$, then both x^* and ω^* are optimal solutions to the primal and dual problems, respectively.*

Strong Duality Theorem:

- *If x^* be the optimal solution of the primal or the dual problem, then there always exist an optimal solution ω^* of the problem such that*

$$C^T x^* = b^T \omega^*,$$

That is the values of the objective functions are same.

- *If the primal problem or the dual problem has unbounded solution, then the other problem always has infeasible solution.*

Ex: Solve the following primal problem:

$$\text{Max } Z = 2x_1 + 3x_2$$

$$\text{Subject to: } 2x_1 + x_2 \leq 4, \quad x_1 + 2x_2 \leq 5, \quad x_1 \geq 0, x_2 \geq 0.$$

Find the solution of the dual of the given primal.

Solution:

		$C_j \rightarrow$	2	3	0	0	Min Ratio $\frac{b}{a_{ij}(>0)}$
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_1	s_2	
0	s_1	4	2	1	1	0	4/1
0	s_2	5	1	2	0	1	5/2 $\rightarrow$
$Z_j - C_j \rightarrow$		$Z = 0$	-2	-3 $\uparrow$	0	0	
0	s_1	3/2	3/2	0	1	-1/2	1 $\rightarrow$
3	x_2	5/2	1/2	1	0	1/2	5
$Z_j - C_j \rightarrow$		$Z = 7.5$	-1/2 $\uparrow$	0	0	3/2	
2	x_1	1	1	0	2/3	-1/3	
3	x_2	2	0	1	-1/3	2/3	
$Z_j - C_j \rightarrow$		$Z = 8$	0	0	1/3	4/3	

The stating variables in the initial table: $\{s_1, s_2\}$.

The corresponding matrix of the variables s_1 and s_2 in final table:

$$B^{-1} = \begin{bmatrix} 2/3 & -1/3 \\ -1/3 & 2/3 \end{bmatrix}.$$

Therefore, the solution of the dual problem:

$$\begin{aligned} [y_1 \ y_2] &= C_B^T B^{-1} = \\ [2 \ 3] \begin{bmatrix} 2/3 & -1/3 \\ -1/3 & 2/3 \end{bmatrix} &= \\ [1/3 \ 4/3]. \end{aligned}$$

That is $y_1 = \frac{1}{3}, \ y_2 = \frac{4}{3},$

$$\text{Min } Z' = 4 \cdot \frac{1}{3} + 5 \cdot \frac{4}{3} = 8$$

Consider the following LP:

$$\text{Maximize } z = x_1 + 5x_2 + 3x_3$$

subject to

$$x_1 + 2x_2 + x_3 = 3$$

$$2x_1 - x_2 = 4$$

$$x_1, x_2, x_3 \geq 0$$

The starting solution consists of x_3 in the first constraint and an artificial x_4 in the second constraint with $M = 100$. The optimal tableau is given as

Basic	x_1	x_2	x_3	x_4	Solution
z	0	2	0	99	5
x_3	1	2.5	1	-.5	1
x_1	0	-.5	0	.5	2

Write the associated dual problem and determine the dual solution from the given table.

The dual of the given problem:

$$\text{Min } Z' = 3y_1 + 4y_2$$

$$\text{Subject to: } y_1 + 2y_2 \geq 1, \quad 2y_1 - y_2 \geq 5, \quad y_1 \geq 3, \quad y_2 \in \mathbb{R}$$

$$\text{Here, } B^{-1} = \begin{bmatrix} 1 & -1/2 \\ 0 & 1/2 \end{bmatrix}, \quad C_B = \begin{bmatrix} 3 \\ 1 \end{bmatrix}. \text{ So, the solution of the dual } [y_1 \quad y_2] = C_B^T B^{-1} = [3 \quad 1] \begin{bmatrix} 1 & -1/2 \\ 0 & 1/2 \end{bmatrix} = [3 \quad -1].$$

That is $y_1 = 3$, $y_2 = -1$ and $\text{Min } Z' = 5$.

Consider the following LP:

$$\text{Maximize } z = 2x_1 + 4x_2 + 4x_3 - 3x_4$$

subject to

$$x_1 + x_2 + x_3 = 4$$

$$x_1 + 4x_2 + x_4 = 8$$

$$x_1, x_2, x_3, x_4 \geq 0$$

Using x_3 and x_4 as starting variables, the optimal tableau is given as

Basic	x_1	x_2	x_3	x_4	Solution
z	2	0	0	3	16
x_3	.75	0	1	-.25	2
x_2	.25	1	0	.25	2

Write the associated dual problem and determine the dual solution from the given table.

The dual of the given problem: $\text{Min } Z' = 4y_1 + 8y_2$

Subject to: $y_1 + y_2 \geq 2$, $2y_1 + 4y_2 \geq 4$, $y_1 \geq 4$, $y_2 \geq -3$, $y_2 \in \mathbb{R}$

Here, $B^{-1} = \begin{bmatrix} 1 & -1/4 \\ 0 & 1/4 \end{bmatrix}$, $C_B = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$. So, the solution of the dual $[y_1 \ y_2] = C_B^T B^{-1} = [4 \ 4] \begin{bmatrix} 1 & -1/4 \\ 0 & 1/4 \end{bmatrix} = [4 \ 0]$.

That is $y_1 = 4$, $y_2 = 0$ and $\text{Min } Z' = 16$.

Complimentary Slackness Conditions:

- a. If in optimal table of primal the decision variable x_k appears as a basic variables then the k –th dual constraint is satisfies as a equality constraint, i.e., slack or surplus variables associated with k –th dual constraints assumes to zero value.
- b. If, in optimal table of primal the slack or surplus variables s_k appears as basic variable then the dual variable y_k associated with the k –th primal constarints assumes zero value in the optimal solution of the dual.

Ex: Solve the following primal problem:

$$\text{Max } Z = 2x_1 + 3x_2$$

$$\text{Subject to: } 2x_1 + x_2 \leq 4, \quad x_1 + 2x_2 \leq 5, \quad x_1 \geq 0, x_2 \geq 0.$$

Find the solution of the dual of the given primal by complimentary slackness conditions.

Solution:

		$C_j \rightarrow$	2	3	0	0	Min Ratio $\frac{b}{a_{ij}(>0)}$
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_1	s_2	
0	s_1	4	2	1	1	0	4/1
0	s_2	5	1	2	0	1	5/2 $\rightarrow$
$Z_j - C_j \rightarrow$		$Z = 0$	-2	-3 $\uparrow$	0	0	
0	s_1	3/2	3/2	0	1	-1/2	1 $\rightarrow$
3	x_2	5/2	1/2	1	0	1/2	5
$Z_j - C_j \rightarrow$		$Z = 7.5$	-1/2 $\uparrow$	0	0	3/2	
2	x_1	1	1	0	2/3	-1/3	
3	x_2	2	0	1	-1/3	2/3	
$Z_j - C_j \rightarrow$		$Z = 8$	0	0	1/3	4/3	

The corresponding dual problem:

$$\text{Min } Z' = 4y_1 + 5y_2$$

$$\begin{aligned} \text{Subject to: } & 2y_1 + y_2 \geq 2, \\ & y_1 + 2y_2 \geq 3, \\ & y_1, y_2 \geq 0. \end{aligned}$$

Since, x_1, x_2 both the decision variable present in the optimal table of the primal problem, so, by CSC the dual solution is given by:

$$\begin{aligned} 2y_1 + y_2 &= 2 \\ y_1 + 2y_2 &= 3 \end{aligned}$$

That is $y_1 = \frac{1}{3}, y_2 = \frac{4}{3}$ and $Z' = 8$.

Consider the following LP:

$$\text{Minimize } z = 4x_1 + x_2$$

subject to

$$3x_1 + x_2 = 30$$

$$4x_1 + 3x_2 \geq 60$$

$$x_1 + 2x_2 \leq 40$$

$$x_1, x_2 \geq 0$$

The starting solution consists of artificial x_4 and x_5 for the first and second constraints and slack x_6 for the third constraint. Using $M = 100$ for the artificial variables, the optimal tableau is given as

Basic	x_1	x_2	x_3	x_4	x_5	x_6	Solution
z	0	0	0	-98.6	-100	-.2	34
x_1	1	0	0	.4	0	-.2	4
x_2	0	1	0	.2	0	.6	18
x_3	0	0	1	1	-1	1	10

Write the associated dual problem, and determine its optimal solution in two ways.

Solution: The corresponding dual problem:

$$\text{Max } Z' = 30y_1 + 60y_2 + 40y_3$$

$$\text{Subject to: } 3y_1 + 4y_2 + y_3 \leq 4,$$

$$y_1 + 3y_2 + 2y_3 \leq 1,$$

$$y_1 \in \mathbb{R}, y_2, y_3 \geq 0.$$

Since, x_1, x_2 both the decision variable present in the optimal table of the primal problem, so, by CSC the dual solution is given by:

$$3y_1 + 4y_2 + y_3 = 4$$

$$y_1 + 3y_2 + 2y_3 = 1 \quad (1)$$

Since, the surplus variable x_3 present in the optimal table, so the corresponding dual $y_2 = 0$.

Then from system (1) reduce to:

$$3y_1 + y_3 = 4$$

$$y_1 + 2y_3 = 1$$

$$\text{That is } y_1 = \frac{7}{5}, y_2 = 0, y_3 = -\frac{1}{5}$$

and $Z' = 8$.

Ex: Consider the following LPP:

$$\text{Max } Z = 5x_1 + 12x_2 + 4x_3$$

Subject to:

$$x_1 + 2x_2 + x_3 \leq 10,$$

$$2x_1 - x_2 + 3x_3 = 8, \quad x_1, x_2, x_3 \geq 0$$

Write the dual of the above problem and using complimentary slackness conditions, find the optimal solution of dual, given that x_1, x_2 are strictly positive in the optimal solution of the primal.

Solution: The dual problem is given by:

$$\text{Min } Z' = 10y_1 + 8y_2$$

Subject to:

$$y_1 + 2y_2 \geq 5$$

$$2y_1 - y_2 \geq 12,$$

$$y_1 + 3y_2 \geq 4, y_1 \geq 0, y_2 \in \mathbb{R}.$$

Standard form $\Rightarrow$

$$\begin{aligned} \text{Min } Z' &= 10y_1 + 8y_2 \\ \text{Subject to: } &y_1 + 2y_2 - s_1 = 5 \\ &2y_1 - y_2 - s_2 = 12, \\ &y_1 + 3y_2 - s_3 = 4. \end{aligned}$$

As given the decision variables x_1 and x_2 present in the optimal table of the primal problem, so the corresponding constraints will be equal to type, $s_1 = 0, s_2 = 0$ i.e.,

$y_1 + 2y_2 = 5$, and $2y_1 - y_2 = 12$, which gives $y_1 = \frac{29}{5}, y_2 = -\frac{2}{5}$.

From the third constraint of dual, we get $s_3 = \frac{3}{5}$.

$$Z' = \frac{274}{5}.$$